CHAPTER

2

Strength of Materials Basics

One of the prerequisites to becoming a pipe stress engineer is being familiar with the principles of strength of materials. In this chapter, some of the basics of this subject relevant to pipe stress engineering will be discussed. The reader should consult standard textbooks on strength of materials for general theories and principles.

Although only the basics of strength of materials are discussed here, some of the items are unique. For instance, the significance of the roundhouse stress-strain curve, the purpose of stress intensity, the meaning of equivalent stress, and so forth are seldom discussed in a standard textbook. Yet, these items are very important in analyzing the stress in piping. This chapter is presented in plain language as much as possible. Therefore, some of the explanations here may not be as rigorous as they would be in an academic setting. The sequence of the discussions will sometimes be in reverse order to what is customarily done in other, more traditional, treatments on the subjects.

2.1 TENSILE STRENGTH

The standard method of measuring the strength of a material is to test its tensile strength by stretching the specimen to failure. Because test results vary considerably with different specimens and procedures, the American Society for Testing and Materials (ASTM) [1] has published a standard for testing and interpretation of results. The test not only determines the ultimate strength of the material but, by stepping up the force gradually, also establishes the relationship between the applied force and the elongation of the specimen.

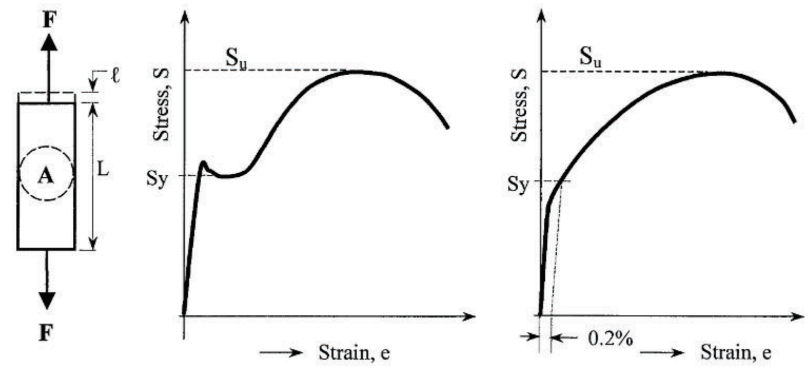
Figure 2.1 shows the relationships between applied force, in terms of stress, and the corresponding elongation produced. At a given stage of the testing, stress and strain are calculated as

$$S = \frac{F}{A} \tag{2.1}$$

$$e = \frac{\ell}{L} \tag{2.2}$$

where

F = applied force
A = cross-section area of the specimen
L = length of the specimen
ℓ = elongation of the specimen



(a), Curve with Yield Point

(b), Round House Curve

FIG. 2.1
BASIC STRESS STRAIN RELATIONS

These two equations are the direct expressions of the following definitions:

Stress (S) is the amount of force per unit cross-section area.
Strain (e) is the amount of elongation per unit length of the specimen.

2.1.1 Modulus of Elasticity

In the elastic range, the stress/strain ratio is constant. This relationship is referred to as Hook's law. The proportional constant is called the modulus of elasticity or Young's modulus, and presented as

$$E = \frac{S}{e} \tag{2.3}$$

The stresses and strains throughout the testing process are plotted in a chart called the stress-strain curve. For piping materials, stress-strain curves can be grouped into two categories. The most familiar one, shown in Fig. 2.1(a), has a pronounced yielding point when an abrupt large elongation is produced without the application of additional force. Most low carbon and low alloy steels have this characteristic. The other category, shown in Fig. 2.1(b), does not have this apparent yield point. The curve runs rather smoothly throughout the process. This type of curve is generally referred to as a roundhouse stress-strain curve. Austenitic stainless steel is one of the most important materials to exhibit this type of stress-strain curve. The stress-strain curve defines the following important design parameters.

2.1.2 Proportional Limit

The curve starts out with a section of straight line. Within this section, the material strictly follows Hook's law, and this portion of the curve is generally referred to as the perfect elastic section. The highest point of this perfect elastic section is called the proportional limit. Young's modulus is defined by this straight section of the curve.

2.1.3 Yield Strength, S_y

The point at which the specimen generates a large deformation without the addition of any load is called the yield point. The corresponding stress is called the yield stress, or yield strength, S_y. The yield point is easy to recognize for materials with a stress-strain curve similar to that shown in Fig. 2.1(a). For materials with a roundhouse stress-strain curve as shown in Fig. 2.1(b), there is no apparent yield point. For these materials, the common approach is to define yield strength by the amount of stress required to produce a fixed amount of permanent deformation. This is the so-called offset method of determining the yield point. ASTM specifies an offset strain of 0.2% to be used for common pipe materials covered by its specifications. This value (0.2%) is quite arbitrary. One rationale is to have yield strength determined in this manner so as to make these values comparable with those of pipe materials with pronounced yield points. It also ensures that the same offset method can be used for materials with pronounced yield points (Fig. 2.1a) and still result in the same yield strength.

Because of this rather arbitrary 0.2% offset definition, the yield strength so determined is not as significant as the real yield strength of materials with pronounced yield points (Fig. 2.1a). This is the reason why some common austenitic stainless steels often have two sets of allowable stresses [2] with different limitations against yield strength. For pressure containing capability, the stress is allowed to reach higher percentage of yield strength if a slightly greater deformation is acceptable. This is mainly because the yield strength of roundhouse materials does not signify an abrupt gross deformation. However, when dealing with seating or sealing, such as with flange connection applications, the 0.2% offset criterion represents sizable damaging deformation. In this case, yield strength is treated similarly to the one with the abrupt yield point, and the allowable stress is limited to the same percentage of the yield strength regardless of material type.

Some piping codes [3] also include a benchmark stress at 1.0% offset to be used as a complement to the 0.2% yield strength. For some materials whose permanent deformation point is difficult to determine, the total deformation may be used. The American Petroleum Institute (API) 5L [4] adopts a 0.5% total elongation to set its yield strength.

2.1.4 Ultimate Strength, S_u

The highest stress on the stress-strain curve is called the ultimate strength. As the material is stretched, the cross-section area will be reduced. However, the stresses given on the curve are determined by dividing the applied force with the original cross-section area of the un-stretched specimen. This explains why the curve shows a drop in stress near the break point toward the end of the curve. The use of the original cross-section area is required as all design calculations are based on original cross-section.

The above tensile test stress is taken at the cross-section plane that is perpendicular to the applied force. In general, stress is not uniform across the whole area. However, with careful arrangement of the specimen shape and test equipment, we can pretty much consider the stress uniform.

2.1.5 Stresses at Skewed Plane

As the force is applied to the specimen, stress is produced not only on the plane perpendicular to the applied force, but also on all other imaginable planes. Figure 2.2 shows the stress generated on the plane that is not perpendicular to the applied force. For simplicity, we consider the specimen to be a small rectangular prism. On an arbitrary plane *m-m* that is inclined with an angle, θ, from the normal plane *m-n*, the applied force, F, decomposes into normal force, F_n, and shear force, F_s. The normal force is perpendicular to the plane and the shear force runs parallel to the plane. The normal force creates the normal stress as given in Eq. (2.4), and the shear force produces the shear stress as given in Eq. (2.5).

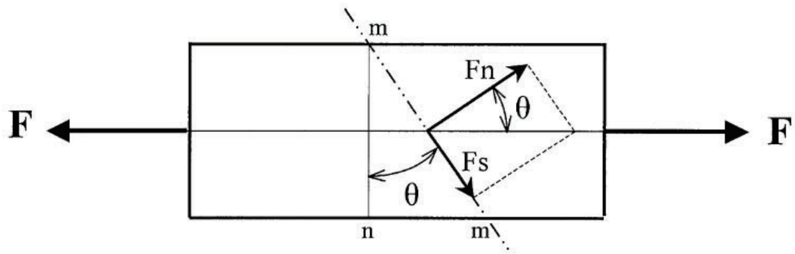


FIG. 2.2
STRESSES AT SKEWED PLANE

$$S_n = \frac{F_n}{A_m} = \frac{F\cos\theta}{A/\cos\theta} = S\cos^2\theta \tag{2.4}$$

$$S_s = \frac{F_s}{A_m} = \frac{F\sin\theta}{A/\cos\theta} = S\sin\theta\cos\theta = \frac{S}{2}\sin 2\theta \tag{2.5}$$

2.1.6 Maximum Shear Stress, $S_{s,max}$

The magnitudes of normal stress and shear stress at a given plane depend on the angle of inclination. For the shear stress, the maximum value is reached when sin 2θ is equal to 1.0. That is, the shear stress is greatest when $2\theta = 90$ deg. or $\theta = 45$ deg. Substituting $\theta = 45$ deg., we have the maximum shear stress equal to one-half of the maximum normal stress, S, as shown in Eq. (2.6)

$$S_{s,max} = \frac{1}{2}S \tag{2.6}$$

The fact that maximum shear stress is one-half of the tensile testing stress leads us to set a stress intensity as twice the value of the maximum shear stress. The stress intensity puts shear stress on the same footing as tensile testing stress and makes them directly comparable to each other. More about the stress intensity will be discussed later in this chapter.

2.1.7 Principal Stresses

Normal stress reaches its maximum level when $\cos^2\theta = 1.0$. This is equivalent to $\theta = 0$ deg. On the other hand, normal stress will be zero, or at its minimum value, when $\theta = 90$ deg. These maximum and minimum normal stresses, which are perpendicular to each other, are called principal stresses. Principal stresses are the basic stresses used in evaluating the damaging effect on the material. The planes on which these principle stresses act upon are called principal planes. There are no shear stresses in principal planes.

2.2 ELASTIC RELATIONSHIP OF STRESS AND STRAIN

The definition of modulus of elasticity as given in Eq. (2.3) is applicable only to the portion of the stress-strain curve that is below the proportional limit. Above the proportional limit, stress and strain do not have a simple mathematical relationship. The stress analysis that follows this constant stress-strain relationship is called elastic analysis. Elastic means that the pipe will return to its original unstressed state when the applied load is completely removed. Most piping stress analyses are either elastic or elastic equivalent analyses. The elastic equivalent analysis uses the elastic method to evaluate the inelastic behavior of the piping. For practical purposes, the stress below the yield strength is generally considered elastic stress.

In elastic analysis, the modulus of elasticity, E, is considered constant throughout the entire stress-strain range. Therefore, in addition to calculating directly from the applied force, the stress can also be calculated from the strain generated. Equation (2.3) can be rearranged for calculating the stress from the strain, and vice versa. From this equation, one might be tempted to calculate the stress by multiplying the modulus of elasticity with all the strain that can be found in the piping. This might lead to an incorrect result, because not all strains are stress-producing strains, as will be discussed in the following section.

2.2.1 Poisson's Ratio

When a specimen is stretched in the x-direction, as shown in Fig. 2.3, a stretching elongation, ℓ_x, is produced in the same direction. This is equivalent to producing a strain of $e_x = \ell_x/L_x$, where L_x is the length of the specimen in the x-direction. While this x-elongation is being created, a measure of shrinkage, ℓ_y, takes place in the y-direction at the same time. This shrinkage at the perpendicular direction to the applied force is a relief effort of the material to maintain a minimum general internal distortion. This phenomenon generates a shrinkage strain of $e_y = -\ell_y/L_y$ in the y-direction. The same thing also happens in the z-direction, which is perpendicular to the paper. The example uses the tensile force as the applied force. A compressive force will develop equivalent strains with all the signs reversed. The absolute value of the ratio of e_y and e_x, resulting from an x-direction force, is a constant called Poisson's ratio.

$$\text{Poisson's ratio, } \nu = \left|\frac{e_y}{e_x}\right| = \frac{\ell_y/L_y}{\ell_x/L_x} \qquad (2.7)$$

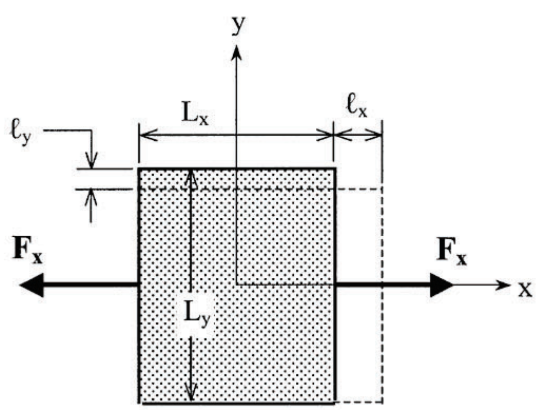


FIG. 2.3
RELIEF (POISSON'S) STRAIN

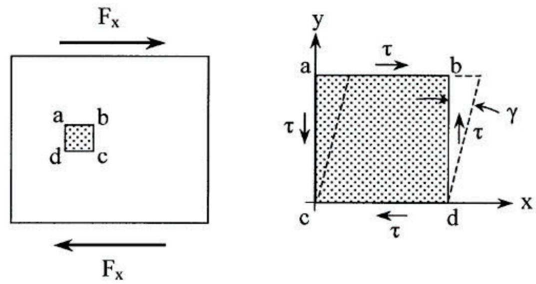


FIG. 2.4
SHEAR DEFORMATION

For metallic piping materials, the Poisson's ratio is roughly equal to 0.3. This is the value used by the American Society of Mechanical Engineers (ASME) B31 Piping Codes [2] when the exact value is not available.

The term e_y, the by-product of the applied strain e_x, is called Poisson's strain or the relief strain. This strain originated from the natural phenomenon of the material. It does not create any stress with its deformation. However, if this deformation is restrained, a stress will be generated. For a general three-dimensional stress situation, the strain in one direction is expressed by the stresses of all three directions as shown in Eq. (2.8).

$$e_x = \frac{S_x}{E} - \nu\frac{S_y}{E} - \nu\frac{S_z}{E}, \quad e_y = \frac{S_y}{E} - \nu\frac{S_z}{E} - \nu\frac{S_x}{E}, \ldots \text{ etc.} \tag{2.8}$$

2.2.2 Shear Strain and Modulus of Rigidity

Shear strain is actually an angular deformation produced by shear stress. Figure 2.4 shows the deformation of an element *a-b-c-d* subject to shear force F_x. The shear stress τ is generated along with a rotation γ. The ratio between them is called the shear modulus of elasticity, or the modulus of rigidity. The modulus of rigidity maintains a constant value within the elastic range. This is generally referred to as an extension of Hook's law. The modulus of rigidity and the modulus of elasticity also have the relation as shown below.

$$G = \frac{\tau}{\gamma}, \quad G = \frac{E}{2(1+\nu)} \tag{2.9}$$

In practical manual piping stress analyses, we seldom have to deal with shear strain. However, shear strain is very important in the implementation of pipe stress analysis computer software, which can easily include the shear deformation that is otherwise too tedious for manual calculations. This distinction between computerized and manual analyses may result in quite different results if the pipe segment is very short. However, the computerized analysis is always more accurate by including the shear deformation.

2.3 STATIC EQUILIBRIUM

To calculate the stress at piping components, the first order of business is to calculate the forces and moments existing at a given cross-section. Calculating these forces and moments at an actual piping system is a very complicated process requiring specialized knowledge and experience. This task nowadays is mostly done with computer software packages developed by analytical specialists. What the practicing stress engineer requires are the basic principles for checking and confirming the results generated by the computer analysis. One of the most important basic principles is that of static equilibrium of forces and moments.

2.3.1 Free-Body Diagram

A free-body diagram is usually used to investigate the internal forces and moments at a given location of the piping component. A free-body is actually a portion of the pipe material enclosed by an arbitrary boundary. However, for the purpose of easy manipulation, the boundary is generally taken to be either a rectangle or a circle. For piping stress analyses, we usually take a section of the pipe, shown in Fig. 2.5, as a free-body. On the boundary of the free-body, we have to include all the forces and moments that exist. For a piping section, there are three directions of forces and three directions of moments at each end of the free-body. In addition, there are body forces that must also be included. The most common body forces are gravity and inertia forces. Figure 2.5 also includes the gravity force, which is equivalent to the weight, *W*, of the body. Inertia force will be discussed in the chapter that deals with dynamic analysis. Surface forces such as wind, friction, and pressure are skipped for the time being.

The forces and moments shown are all acting on the body. Those at the ends are internal forces and moments that are common to the bodies sharing the same boundary. For instance, the forces and moments at end-*b* of this *a-b* body are the same forces and moments acting on the end-*b* of the *b-c* body, except that the directions are reversed.

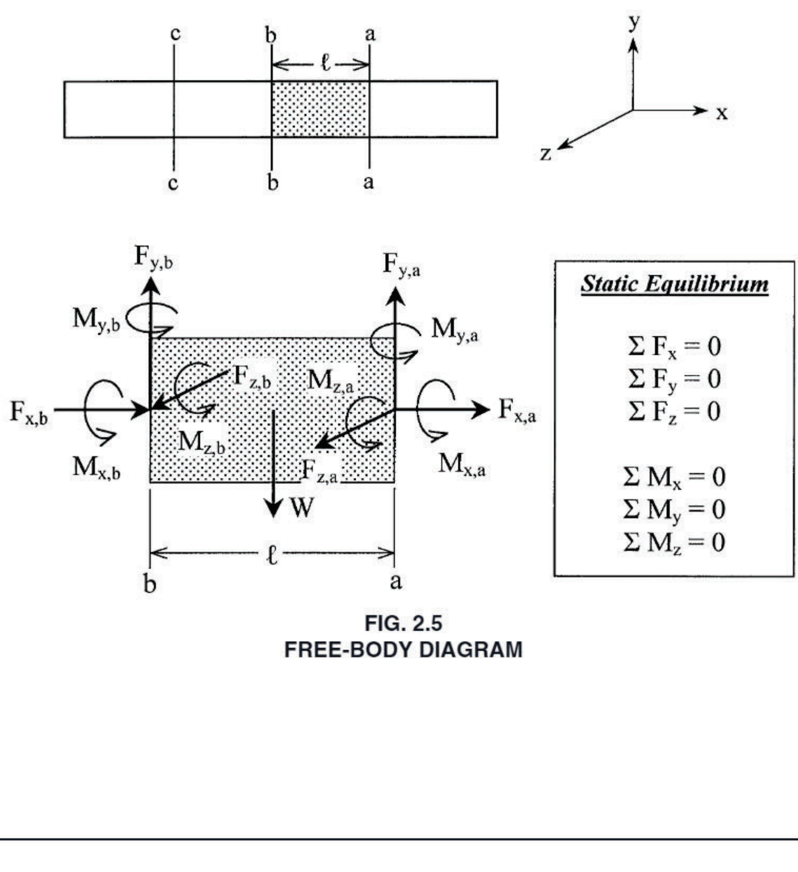


FIG. 2.5
FREE-BODY DIAGRAM

2.3.2 Static Equilibrium

To satisfy the condition that the body should not move ceaselessly, the resultant forces and moments of the entire body have to be zero (balance). This is the main principle of static equilibrium. Mathematically, we are used to calling it $\Sigma F_x = 0$, $\Sigma F_y = 0,\ldots$, $\Sigma M_z = 0,\ldots$, and so forth. Verbally, it states that the summation of forces in each direction is zero, and the summation of moments in each direction is also zero. The summation of forces is straightforward; however, summing the moments is quite different. In moments, we are talking about the summation of moment effects rather than the algebraic summation of the apparent moments. These equilibrium relations are often used to calculate the forces and moments at a given location from the known forces and moments at another location. For instance, if we know the forces and moments at end-a, then the forces and moments at end-b can be determined by the following relations:

(a) Forces: They are calculated using the following straightforward algebraic summations. Note that weight, W, is the gravity force acting toward the negative y direction.

$$\begin{aligned}
\Sigma F_x &= 0: \quad F_{x,a} + F_{x,b} = 0; && \text{or } F_{x,b} = -F_{x,a} \\
\Sigma F_y &= 0: \quad F_{y,a} + F_{y,b} - W = 0; && \text{or } F_{y,b} = -F_{y,a} + W \\
\Sigma F_z &= 0: \quad F_{z,a} + F_{z,b} = 0; && \text{or } F_{z,b} = -F_{z,a}
\end{aligned}$$

(b) Moments: Because the total moment effect about any given point has to be zero, any point in the body can be chosen to check the equilibrium relation. A point with expected unknown forces is normally chosen for this purpose to simplify the calculation. The following are calculated by taking the moment about point b.

$$\begin{aligned}
\Sigma M_x &= 0: \quad M_{x,a} + M_{x,b} = 0; && \text{or } M_{x,b} = -M_{x,a} \\
\Sigma M_y &= 0: \quad M_{y,a} + M_{y,b} - F_{z,a}\ell = 0; && \text{or } M_{y,b} = -M_{y,a} + F_{z,a}\ell \\
\Sigma M_z &= 0: \quad M_{z,a} + M_{z,b} + F_{y,a}\ell - W\ell/2 = 0 && \text{or } M_{z,b} = -M_{z,a} - F_{y,a}\ell + W\ell/2
\end{aligned}$$

In the descriptions of piping stress analysis topics, the generic term "force" implies both force and moment. By the same token, the generic term "displacement" implies both displacement and rotation.

2.4 STRESSES DUE TO MOMENTS

The stress at any point on a given plane of the material is calculated by dividing the force with the area as defined in Eq. (2.1). The equation is applicable only to uniform stress distributions. For a non-uniform stress distribution, a very small area is considered at each given point for the purpose of calculating the stress at that point.

When a force is applied on a solid bar of practical size, we can pretty much assume that the stress is uniform and calculate the stress by dividing the applied force with the cross-sectional area. However, for moment loads, the magnitude of the stresses changes across the entire cross-section, thus requiring different treatment. There are two types of moment loads: bending moment and torsion (or twisting) moment. Assuming that the centerline of the pipe element is lying in the x-direction, then the bending moments, M_y and M_z, bend the pipe in the lateral directions, whereas the torsion moment, M_x, twists the pipe in the axial direction. Bending and twisting produce different types of stresses and stress distributions. The stresses due to bending moments will be discussed first.

2.4.1 Stresses due to Bending Moments

Figure 2.6 shows a moment, M, acting on a section of a beam, which has a rectangular cross-section of h by b. The moment has the tendency of bending the beam into an arc shape. The basic beam

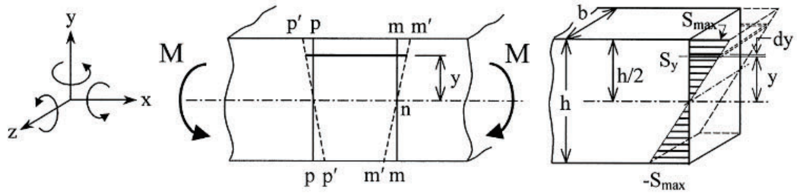


FIG. 2.6
BENDING STRESS DISTRIBUTION

formula, based on experience, assumes that a transverse plane of the beam remains plane and normal to the longitudinal fibers of the beam after bending. For instance, planes *m-m* and *p-p* will remain straight planes, *m′-m′* and *p′-p′*, after bending. The stress created is proportional to the deformation represented by the movement *m-m′*. With the moment direction as shown, tensile stress is generated at the top portion, and compressive stress is generated at the bottom portion of the cross-section. By further assuming that the material has the same tensile and compressive characteristics, it becomes apparent that the identical distribution of tensile stress and compressive stress is required to balance the axial forces. With identical tensile and compressive stress distributions, a zero stress is established at the center surface. The surface of zero stress is called the neutral surface, and the intersection of the neutral surface with any cross-section is called the neutral axis.

The bending stress at the beam is calculated by equalizing the moment generated by the stress with the moment applied. In Fig. 2.6, assuming that the maximum stress at the outer fiber is S_{max}, then the stress at y distance away from the neutral axis is $S_y = S_{max}\,y/(h/2) = (2S_{max}\,y)/h$. Taking the moment about the Z-neutral axis, the moment generated by the stress consists of two equal parts, one due to tensile stress and the other due to compressive stress. The sum of the moments is, therefore, twice the moment generated by the tensile stress. That is,

$$M = 2\int_0^{h/2} bS_y * y * \mathrm{d}y = 2\int_0^{h/2} b(2S_{max}/h)y^2\mathrm{d}y$$

$$= \frac{2S_{max}}{h} * 2\int_0^{h/2} b * y^2\mathrm{d}y = \frac{2S_{max}}{h} * I$$

$$S_{max} = M\frac{h}{2I} = \frac{M}{Z} \tag{2.10}$$

where

$$I = 2\int_0^{h/2} b * y^2\mathrm{d}y = 2b\left[\frac{y^3}{3}\right]_0^{h/2} = \frac{bh^3}{12}$$

is the moment of inertia, about the z-axis, of the rectangular beam cross-section. And

$$Z = \frac{I}{h/2} = \frac{bh^2}{6}$$

is the corresponding section modulus.

In piping stress analysis, most common cross-sections are circular. Equation (2.10) is also applicable to a circular cross-section, using a different moment of inertia and section modulus. For a solid circular cross-section of diameter D, the moment of inertia and section modulus are [5, 6]:

$$I = \frac{\pi D^4}{64}, \quad Z = \frac{\pi D^3}{32}$$

These formulas can also be easily derived from the polar moment of inertia to be discussed later in this section.

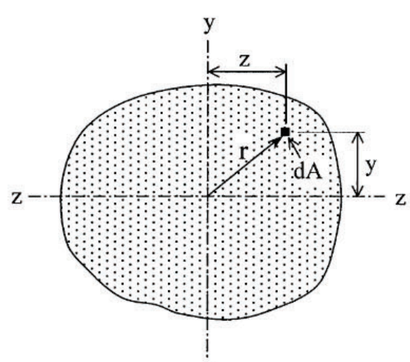
The maximum bending stress calculated is located at the extreme outer fiber of the cross-section. Stresses at the inner locations are smaller and are proportional to the distance from the neutral axis. Because of this localized nature of the maximum stress, the damaging effect of bending stress is considered to be not as severe as a uniform stress of the same magnitude.

2.4.2 Moment of Inertia

The moment of inertia and section modulus have been introduced in the calculation of bending stress. The moment of inertia about a given axis is formally defined as the sum of the products of each elementary area of the cross-section multiplied by the square of the distance from the area to the axis. Figure 2.7 shows the definitions and formulas for the moment of inertia, $I_{z\text{-}z}$ and $I_{y\text{-}y}$, around *z-z* and *y-y* axes, respectively. $I_{z\text{-}z}$ resists the M_z moment that is acting around the *z-z* axis, whereas $I_{y\text{-}y}$ resists the M_y moment acting around the *y-y* axis. For a symmetrical cross-section, areas with the same moment arm are lumped together for easier integration. In a rectangular cross-section (Fig. 2.6), we have lumped the area with the same moment arm, y, to form a strip of width b. The summation is then only required to integrate these strips starting from the neutral axis to the outer surface.

2.4.3 Polar Moment of Inertia

With twisting moments, we are dealing with the polar rotation arm, r, rather than the cranking arms, z and y, used in the bending moment of inertia. The moment of inertia determined by the polar arm is



$$I_{z-z} = \int_A y^2 dA$$

$$I_{y-y} = \int_A z^2 dA$$

$$\begin{aligned} I_p &= \int_A r^2 dA \\ &= \int_A (y^2 + z^2) dA \\ &= I_{z\text{-}z} + I_{y-y} \end{aligned}$$

FIG. 2.7
MOMENT OF INERTIA OF CROSS SECTION

called the polar moment of inertia, I_p. Because $r^2 = z^2 + y^2$, the polar moment of inertia is the sum of the two bending moment of inertias about the two mutually perpendicular axes. This is given as

$$I_{z\text{-}z} = \int_A y^2 \mathrm{d}A, \qquad I_{y\text{-}y} = \int_A z^2 \mathrm{d}A$$
$$I_p = \int_A r^2 \mathrm{d}A = \int_A (y^2 + z^2)\mathrm{d}A = I_{z\text{-}z} + I_{y\text{-}y} \tag{2.11}$$

Equation (2.11) is often used to determine the polar moment of inertia by calculating the bending moment of inertia. It is also a convenient tool for calculating the moment of inertia of a circular cross-section from the polar moment of inertia. For a circular cross-section, the calculation of the moment of inertia is quite cumbersome, but the calculation of the polar moment of inertia around the center point is fairly straightforward. Because $I_{y\text{-}y}$ and $I_{z\text{-}z}$ are equal for circular cross-section, from Eq. (2.11) we have $I_{y\text{-}y} = I_{z\text{-}z} = I_p/2$. The polar moment of inertia is calculated by integrating the concentric rings, $2\pi r\,\mathrm{d}r$, as follows:

$$I_p = \int_0^R r^2 * 2\pi r * \mathrm{d}r = 2\pi \int_0^R r^3 \mathrm{d}r = 2\pi \left[\frac{r^4}{4}\right]_0^R = \pi \frac{R^4}{2} = \frac{\pi D^4}{32}$$

$$Z_p = \frac{I_p}{R} = \frac{\pi D^3}{16}$$

2.4.4 Moment of Inertia for Circular Cross-Sections

The preceding polar moment of inertia can be used to calculate the moment of inertia of circular cross-sections as

$$I = I_{z\text{-}z} = I_{y\text{-}y} = \frac{I_p}{2} = \frac{\pi R^4}{4} = \frac{\pi D^4}{64}$$

$$Z = \frac{I}{R} = \frac{\pi D^3}{32}$$

2.4.5 Stresses due to Torsion Moment

The twisting of a bar with a non-circular cross-section is a very complicated process. In this book, only the twisting of a bar with a circular cross-section will be discussed. As shown in Fig. 2.8, when a circular bar is twisted by the torsion moment, M_t, a rectangular element *a-b-c-d* will be deformed into a skewed parallelogram. This angular deformation produces shear stress in response to the torsion moment.

When calculating the torsion shear stress, the circular cross-section is assumed to remain circular and the plane perpendicular to the axis remains perpendicular to the axis, after the twisting. This intuitive assumption, based on the all-around symmetric nature of the circular cross-section, is also verified by experiments. Based on this assumption, the stress on the plane perpendicular to the axis is pure shear without any axial component. For the rectangular element *a-b-c-d*, the shear stress is proportional to the angular deformation shown in the dotted lines. This angular deformation diminishes toward the centerline of the bar and reduces to zero at the center. Assuming that the maximum stress at the surface is τ_{max}, then the stress at the location r distance from the center is $\tau = \tau_{max} r/R$. The shear stress can be calculated by the equilibrium of the moment generated by the stress and the moment applied. That is,

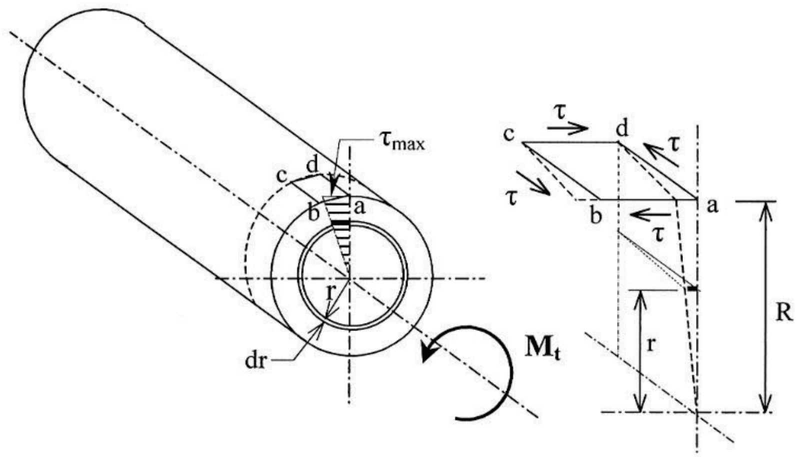


FIG. 2.8
STRESSES DUE TO TORSION MOMENT

$$M_t = \int_0^R \tau * 2\pi r dr * r = \int_0^R \frac{\tau_{max} r}{R} * 2\pi r dr * r = \frac{\tau_{max}}{R} \int_0^R 2\pi r dr * r^2$$
$$= \frac{\tau_{max}}{R} * I_p$$

$$\tau_{max} = M_t \frac{R}{I_p} = \frac{M_t}{Z_p} \tag{2.12}$$

where

I_p = polar moment of inertia of the cross-section
$Z_p = I_p/R$ = torsion section modulus

Equation (2.12) is analogous to Equation (2.10), except that the value of Z_p is exactly twice that of Z for circular cross-sections such as pipe cross-sections.

Figure 2.8 also shows that with pure shear stress, τ, on face *a-b*, there will also be a shear stress of the same magnitude in the perpendicular face *b-c*. The directions of these shear stresses are so determined that the net twisting effect of the element is balanced. Because a non-circular cross-section does not have the capability to maintain this perpendicular shear stress at the corners, a warping will be initiated at the corner surfaces. Due to warping of the cross-section, a plane perpendicular to the axis does not remain perpendicular after the twisting; Eq. (2.12) is not applicable to bars with non-circular cross-sections. Outside these *a-b-c-d-a* surfaces, where only shear stress exists, are other surfaces with normal stresses. At 45 deg. from the *a-b* plane, the shear stress is zero and the normal stress is at maximum. These maximum normal stresses have the same magnitude as the maximum shear stress. They are tension at one plane and compression at the perpendicular plane.

2.5 STRESSES IN PIPES

A piping component experiences two main categories of stresses. The first category of stress comes from the pressure, either internal or external. The second category of stress comes from the forces and moments generated by weight, thermal expansion, wind, earthquake, and so forth.

2.5.1 Stresses due to Internal Pressure

The most common and important stress at a piping component is the stress due to internal pressure. When a pipe is pressurized, its inside surface is exposed to the same pressure in all directions. The pressure force is acting in the normal direction of the surface. However, we generally do not have to deal with every detail of the internal surface to determine the effect of the pressure and to calculate the stress due to pressure. Figure 2.9 shows a leg of piping spool subject to an internal pressure, *P*. Because of this pressure, the pipe wall is stretched in every direction. From the symmetry of the circular cross-section, we intuitively assume that there are two principal stresses, axial and circumferential, developed uniformly along the circumference of the pipe wall. These two stresses acting on a typical pipe wall element are designated as S_{lp} and S_{hp}, respectively, as shown in Fig. 2.9(a). The stress in the axial direction is called the longitudinal pressure stress, and the one in circumferential direction is called the hoop pressure stress. To calculate the magnitude of these stresses, a ring *m-m-n-n* containing the element is taken as the free-body. The stresses are then calculated by the equilibrium of the pressure force and the stress force acting at the boundaries.

Longitudinal pressure stress, S_{lp}. The longitudinal stress on the pipe is generally considered uniform in both circumferential and diametrical locations. Because it is required to have uniform longitudinal strain across the cross-section and across the thickness to prevent the pipe from being flared into a funnel shape, elastically, the longitudinal stress varies slightly with the circumferential stress due to Poisson's effect. However, a uniform longitudinal stress distribution is considered for both straight and bend sections in practical applications. From Fig. 2.9(b), we know that the longitudinal forces

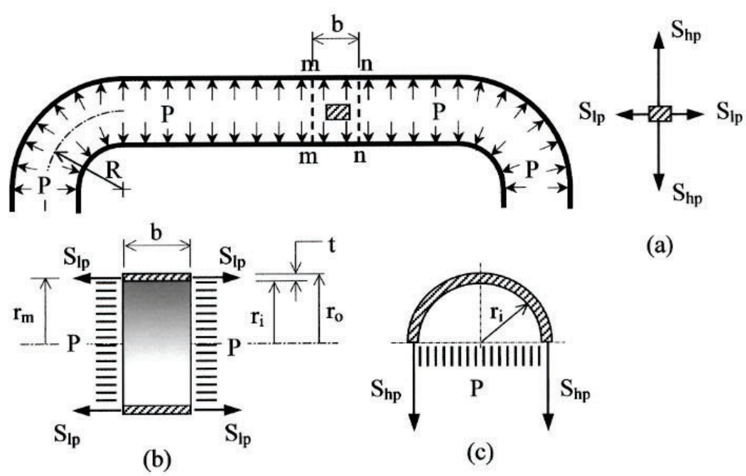


FIG. 2.9
STRESSES DUE TO INTERNAL PRESSURE

at both sides of the free-body are identical. Therefore, the forces balance out on each side of the free-body. By taking the equilibrium of the forces in the axial direction over the entire circular cross-section, we have

$$\text{Pressure force} = \pi r_i^2 P$$
$$\text{Stress force} = \pi(r_o^2 - r_i^2)\, S_{lp} = \pi(r_o - r_i)(r_o + r_i)\, S_{lp} = \pi t(2r_m)\, S_{lp}$$

Because the stress force balances the pressure force, we have

$$r_i^2 P = t(2r_m)\, S_{lp}$$

$$S_{lp} = \frac{r_i^2 P}{2r_m t} < \frac{r_m^2 P}{2r_m t} = \frac{r_m P}{2t} < \frac{r_o P}{2t} \tag{2.13}$$

Equation (2.13) represents several variations of the formulas that might appear in different articles on this subject. The first expression is the most accurate one, whereas the last expression, $r_oP/2t$, is the most conservative one yielding the highest stress.

Hoop pressure stress, S_{hp}. The distribution or variation of the hoop pressure stress is not as uniform as the longitudinal stress. In the diametrical direction, the stress is higher at the inner surface and lower at the outside surface. The piping code [2] has given a design formula that properly takes into account this non-uniform stress distribution. This section will only discuss the general simplified formula that assumes that stress is uniform across the thickness. The code requirements will be discussed later in a separate chapter.

By balancing the forces acting on the semi-circular band as given in Fig. 2.9(c) for a band width, b, we have

$$\text{Pressure force} = 2r_i bP, \text{ which should be equal to}$$
$$\text{Stress force} = 2bt * S_{hp}$$
$$\text{or } S_{hp} = \frac{r_i}{t}P$$

The above relation assumes a uniform hoop stress across the thickness. In reality, the stress is not uniform and is greater near the inside surface. To compensate for this non-uniform stress distribution, the design equation normally uses the outside radius instead of the inside radius as

$$S_{hp} = \frac{r_o}{t}P \tag{2.14}$$

From Eqs. (2.13) and (2.14), it is clear that hoop pressure stress is roughly twice the value of longitudinal pressure stress.

Hoop pressure stress at a bend of uniform thickness has some variation across the circumference. If a radial strip is taken as a free-body, it is clear that the inner curvature (crotch) area has less area to resist the pressure force compared to the outer curvature area. Therefore, a higher hoop stress is expected at the crotch area. The theoretical hoop stresses at a bend of uniform thickness can be calculated by multiplying the hoop stress of the straight pipe as given by Eq. (2.14) with the following factors (Derivations of these factors are given in Chapter 4 where the pressure design of piping components is discussed.):

$$\frac{(2R - r_m)}{(2R - 2r_m)} \cong \frac{4(R/D) - 1}{4(R/D) - 2} \quad \text{at intrados (crotch)}$$
$$\frac{(2R + r_m)}{(2R + 2r_m)} \cong \frac{4(R/D) + 1}{4(R/D) + 2} \quad \text{at extrados (crown)}$$

These factors, occasionally called the Lorenz factors, are sometimes used as stress intensification factors due to pressure at the bends. They are also used to gauge the actual wall thickness required at different locations of a bend. However, in most practical applications, they tend to be ignored. This is partially because a forged or hot rolled bend generally has a thicker wall at the crotch area, thus neutralizing the effect of these factors. The safety factor normally included in the design code and specification also covers this type of minor deviations.

2.5.2 Stresses due to Forces and Moments

Besides the pressure that generates the pressure stress discussed in the preceding subsection, other loadings can also produce internal forces and moments that generate significant stress in the pipe. These internal forces and moments acting at a given pipe cross-section (Fig. 2.10) are the result of thermal expansion, weight, wind, earthquake, and other internal and external loads applied to the piping system.

Stresses due to forces. In a piping component, forces can be divided into two categories: shear force, F_s, which acts in a direction perpendicular to the pipe axis; and axial force, F_a, which acts in the axial direction of the pipe. With coordinate axes selected as in Fig. 2.10, the shear force comprises two forces, F_y and F_z, each of which produces a shear stress at the pipe cross-section. The stress is not uniform, and is greatest at the diametrical centerline perpendicular to the force. The ratio of the maximum value and the average value is called the shear distribution factor. For most pipe cross-sections, the shear distribution factor is very close to 2.0. Therefore, we can write

$$\tau_{xy,\max} = 2\frac{F_y}{A}, \quad \tau_{xz,\max} = 2\frac{F_z}{A}, \quad A = \pi(r_o^2 - r_i^2) = 2\pi r_m t$$

τ_{xy} implies that the shear stress is parallel to the y-axis and acting on the plane perpendicular to the x-axis. Normally, we are only interested in the combined maximum shear stress at the cross-section.

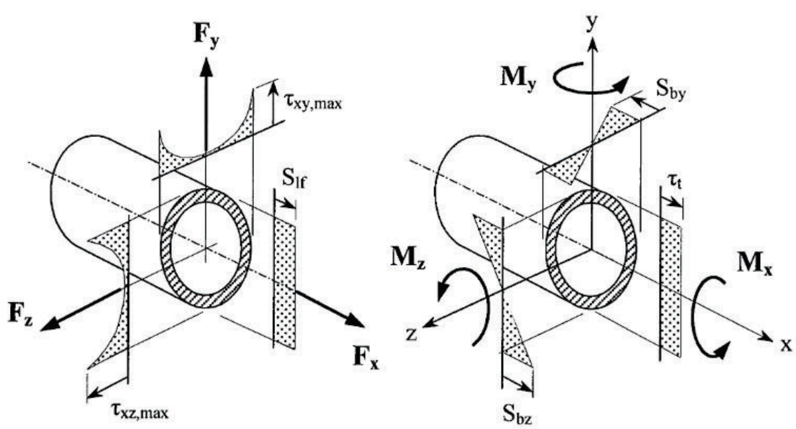


FIG. 2.10
STRESSES DUE TO FORCES AND MOMENTS

In this case, the two shear forces are first combined into a resultant shear force as $F_s = \sqrt{F_y^2 + F_z^2}$. The stress is then calculated using this resultant force.

For the axial force, $F_a = F_x$, the stress is either pure tension or compression, depending on the sign of the force. The axial stress generated is uniform across the cross-section. Therefore, at a given cross-section of the pipe we have the stresses due to forces as

$$S_{lf} = \frac{F_a}{A}, \quad \tau_f = 2\frac{F_s}{A}, \quad A = 2\pi r_m t \tag{2.15}$$

where

S_{lf} = longitudinal stress due to the force
τ_f = shear stress due to the force

Both stresses are acting on the same cross-section plane. Stresses due to direct forces as given in Eq. (2.15) are generally insignificant when compared with stresses due to moments. Therefore, stresses due to direct forces are often ignored in the evaluation of the piping system.

Stresses due to moments. Moment loads are also divided into two categories: bending moment and torsion moment. The bending moment is further divided into two components, M_y and M_z, around the conveniently selected y-axis and z-axis, respectively. As discussed in Section 2.4, each bending moment creates a linear distribution of stress with the highest stress occurring at the outer surface farthest from the bending axis and is equal to

$$S_{by} = \frac{M_y}{Z}, \quad S_{bz} = \frac{M_z}{Z}, \quad Z = \frac{\pi}{4r_o}(r_o^4 - r_i^4)$$

S_{by} and S_{bz} are located at the outside surface of the pipe, but are 90 deg. apart from each other. They are normally combined together to become the total bending stress. That is,

$$S_b = \sqrt{S_{by}^2 + S_{bz}^2} = \frac{1}{Z}\sqrt{M_y^2 + M_z^2}$$

The stress created by the torsion moment, $M_t = M_x$, is shear stress that is linearly distributed in the diametrical direction with the maximum at the outer surface. The shear stress, however, is uniformly distributed along the circumferential direction. The magnitude of the highest shear stress is calculated by

$$\tau_t = \frac{M_t}{Z_p} = \frac{M_t}{2Z}$$

where τ_t is shear stress due to the torsion moment. This stress needs to be combined with bending and other stresses to evaluate the piping system.

For pipes with thin walls, the formulas for the section modulus can be simplified. In a thin wall pipe, we can assume that the cross-section area is concentrated to a ring having a radius of r_m. Then, from the definition of the polar moment of inertia, we have the following

$$I_p = 2\pi r_m t * r_m^2 = 2\pi r_m^3 t, \quad Z_p = \frac{I_p}{r_m} = 2\pi r_m^2 t, \quad Z = \frac{Z_p}{2} = \pi r_m^2 t$$

These approximate formulas appear occasionally in the code book [2]. The preceding lays the background.

2.6 EVALUATION OF MULTI-DIMENSIONAL STRESSES

A piping component is generally exposed to multi-axial stresses. The evaluation of these stresses is not as simple as our intuition may lead us to think. For instance, Fig. 2.11 shows a few combinations of three directional principal stresses. The absolute magnitudes of the stresses are the same. The body in case (a) has all three directions subject to tension, in case (b) all in compression, in case (c) with one direction in compression, and in case (d) with one direction at zero stress. Intuitively, case (b) appears safest to us because all the stresses are trying to keep the body together. This, in fact, proves to be the case as proved by experiments on a body subjected to a large external hydraulic pressure. However, our assessments of the other cases are not as clear. For instance, we may think case (a) is bad, because all the stresses have the tendency of tearing the body apart. But in reality, it is as good as case (b), except that it is hard to test in real life. How about case (c), where one direction of stress is in compression? This is where our intuition fails to reach the right conclusion. This may surprise us, but case (c) is actually considerably worse than case (a) in terms of safety. In case (d), one direction of tensile stress is removed. This case (d) fares just as bad as, but slightly better than, case (c). As will be derived later in this section, damage to the body is determined by the difference of the maximum and minimum principal stresses.

Presentation of a general three-dimensional stress field is very tedious and complicated. In the following, we will discuss the general two-dimensional stress field to get acquainted with the general philosophy of the code stress evaluation. A three-dimensional stress field can generally be evaluated using the two-dimensional approach, investigating one plane at a time.

2.6.1 General Two-Dimensional Stress Field

Figure 2.12 shows a general two-dimensional stress field on the piping component. The dimension in the direction perpendicular to the paper is immaterial and can be considered the thickness of the pipe in most cases. In this general stress field, there are normal stress and shear stress, as denoted, at each side of the rectangular free-body. For normal stress, tension is considered positive and compression negative. The sign convention for shear stress is not as well defined as for normal stress. Fortunately, this ambiguity of sign in shear stress does not materially affect the outcome of our discussion. The following discussion and derivation are based on the assumption that signs are positive as shown. The body is assumed to align with the x and y coordinates as shown.

The first step of the evaluation requires the calculation of maximum normal stress and shear stress that might exist at any plane of the body. This shall start with the determination of the normal and shear stress at any plane m-m located with a θ angle from the x-axis. From the stresses on this general plane, the maximum stresses from all conceivable planes can be determined. To find the stresses at

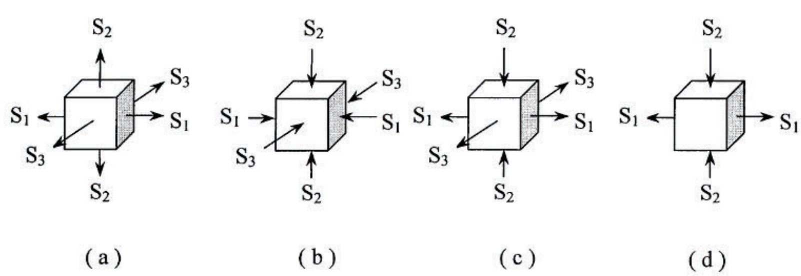


FIG. 2.11
ELEMENTS SUBJECT TO THREE-DIMENSIONAL STRESSES

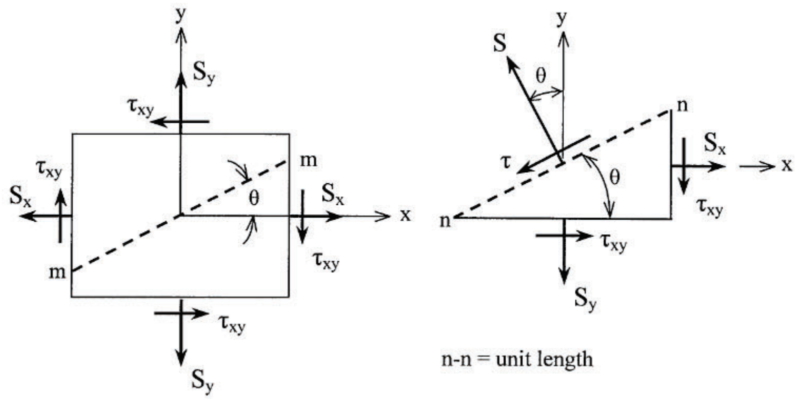


FIG. 2.12
TWO-DIMENSIONAL STRESS RELATIONS

plane *m-m*, a right triangle free-body with *m-m* as the cord is constructed. Taking a unit length *n-n* along the *m-m* plane, we have the base equal to cos θ and the height equal to sin θ. By balancing the forces along the normal and parallel directions to the plane, we have the following relations:

$$\begin{aligned} S &= S_x \sin\theta \sin\theta + S_y \cos\theta \cos\theta + \tau_{xy} \sin\theta \cos\theta + \tau_{xy} \cos\theta \sin\theta \\ &= S_x(1-\cos 2\theta)/2 + S_y(1+\cos 2\theta)/2 + \tau_{xy} \sin 2\theta \\ &= \frac{1}{2}(S_x + S_y) - \frac{1}{2}(S_x - S_y)\cos 2\theta + \tau_{xy}\sin 2\theta \end{aligned} \tag{2.16a}$$

$$\begin{aligned} \tau &= S_x \sin\theta \cos\theta - S_y \cos\theta \sin\theta - \tau_{xy} \sin\theta \sin\theta + \tau_{xy} \cos\theta \cos\theta \\ &= \frac{1}{2}(S_x - S_y)\sin 2\theta + \tau_{xy}\cos 2\theta \end{aligned} \tag{2.16b}$$

The θ of the plane where the maximum and minimum normal stresses occur can be found by differentiating Eq. (2.16a) with respect to θ, and setting the derivative to zero. Maximum stress is then calculated by back-substituting this θ into the equation. Maximum shear stress can also be determined in the same manner. However, an elegant pictorial representation can be developed by first moving $(S_x + S_y)/2$ term in Eq. (21.6a) to the left-hand side, and then squaring both (2.16a) and (2.16b) and summing them together as follows:

$$\begin{aligned} \left[S - \frac{1}{2}(S_x + S_y)\right]^2 + \tau^2 &= \left[\frac{1}{2}(S_x - S_y)\cos 2\theta - \tau_{xy}\sin 2\theta\right]^2 \\ &\quad + \left[\frac{1}{2}(S_x - S_y)\sin 2\theta + \tau_{xy}\cos 2\theta\right]^2 \end{aligned}$$

or

$$\left[S - \frac{1}{2}(S_x + S_y)\right]^2 + \tau^2 = \left(\frac{S_x - S_y}{2}\right)^2 + \tau_{xy}^2 \tag{2.16c}$$

2.6.2 Mohr's Circle for Combined Stresses

Equation (2.16c) is a general equation of a circle in S-τ coordinates. It shows that the relationship between normal stress, S, and shear stress, τ, at any given plane can be expressed with a circle. The center of the circle is located at $(S_x + S_y)/2$ away from the origin on the S-axis, and the radius of the circle is the square root of the right-hand side of Eq. (2.16c). This relationship can be expressed with Mohr's stress circle as shown in Fig. 2.13. The procedure for drawing the circle is as follows: (1) Draw the horizontal axis as the S coordinate and the vertical axis as the τ coordinate. (2) Locate S_x and S_y on the S coordinate. Here, S_x is assumed to be greater than S_y. (3) At S_x or S_y points, draw a vertical line that is equal to τ_{xy}. (4) Locate the mid-point between S_x and S_y to serve as the center point, and draw a circle with the distance between the center point and the τ_{xy} point defined in (3) as the radius.

From the figure and the equation, the maximum and minimum normal stresses are found by setting $\tau = 0$, and taking the square roots of both sides of the equation. The results are as follows:

$$S_1 = \frac{S_x + S_y}{2} + \sqrt{\left(\frac{S_x - S_y}{2}\right)^2 + \tau_{xy}^2} \tag{2.17a}$$

$$S_2 = \frac{S_x + S_y}{2} - \sqrt{\left(\frac{S_x - S_y}{2}\right)^2 + \tau_{xy}^2} \tag{2.17b}$$

The maximum shear stress is equal to the radius of the circle. That is,

$$\tau_{\max} = \sqrt{\left(\frac{S_x - S_y}{2}\right)^2 + \tau_{xy}^2} = \frac{S_1 - S_2}{2} \tag{2.18}$$

The significance of the combined stress is determined by its potential for damaging the structure. Its assessment would require the understanding of the theory of failure. For this discussion, the general three-dimensional body with three principal stresses S_1, S_2, and S_3 related by $S_1 > S_2 > S_3$ is used as an example. Compressive stresses are regarded as negative stresses.

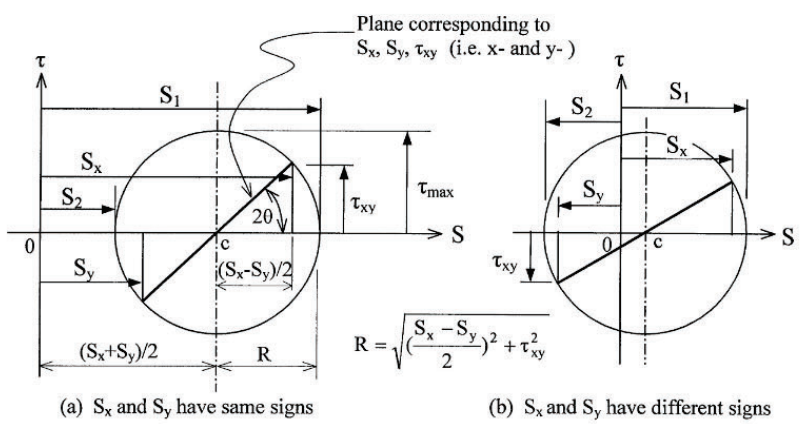


FIG. 2.13
MOHR'S STRESS CIRCLE FOR TWO-DIMENSIONAL STRESSES

2.6.3 Theories of Failure

There are several different theories of failure that have been proposed and used [7]. These theories include: (1) *maximum stress theory*, which predicts that the material will yield when the absolute magnitude of any of the principal stresses reaches the yield strength of the material; (2) *maximum strain theory*, which predicts that the material will yield when the maximum strain reaches the yield point strain; (3) *maximum shear theory*, which predicts that the material will yield when the maximum shear stress in the material reaches the maximum shear stress at the yield point in the tension test; (4) *maximum energy theory*, which predicts that the material will yield when the strain energy per unit volume in the material reaches the strain energy per unit volume at yielding in the simple tension test; and (5) *maximum distortion energy theory*, which predicts that the material will yield when the distortion energy per unit volume in the material reaches the distortion energy per unit volume at yielding in simple tension test.

The maximum stress theory fits very well with brittle materials such as concrete and non-ductile cast iron. For the ductile materials that are prevalent in piping, both the maximum shear theory and the maximum distortion energy theory agree very well with the experiments. The maximum distortion energy theory is slightly more accurate, but the maximum shear theory is simpler and easier to apply. ASME has adopted the maximum shear failure theory in its piping and pressure vessel codes.

2.6.4 Stress Intensity (Tresca Stress)

In the maximum shear failure theory, the condition for yielding occurs when the maximum shear stress in the material equals the maximum shear stress at the yield point in the tension test. From Eq. (2.6), the maximum shear stress at the yield point in the tension test is equal to one-half of the tensile yield strength. In other words, twice the maximum shear stress at yield in the tension test is the same as the tensile yield strength. Therefore, by *defining stress intensity as twice the maximum shear stress*, it can then be directly compared to the tensile yield strength and other data obtained in the tension test. In Eq. (2.18), for a two-dimensional stress field, we have

$$\text{Stress intensity, } S_i = 2\tau_{max} = S_1 - S_2$$

The above expression can be extended to a three-dimensional stress field as $S_i = S_1 - S_3$, where $S_1 > S_2 > S_3$. Compressive stresses are considered negative stresses in the above comparative quantity. This provides the basis for the comparative damaging effect of the stress systems shown in Fig. 2.11.

In piping stress analysis, we deal mostly with two-dimensional stress fields with the stress at the third dimension either zero or insignificant. In this case, it is simpler to calculate the stress intensity directly from the general stress field given in Fig. 2.12, without calculating the principal stresses. Again, from Eq. (2.18) we have

$$S_i = 2\tau_{max} = 2\sqrt{\left(\frac{S_x - S_y}{2}\right)^2 + \tau_{xy}^2} = \sqrt{(S_x - S_y)^2 + 4\tau_{xy}^2} \tag{2.19}$$

Assuming that x-axis is in the axial direction of the pipe, then S_x is the longitudinal stress resulting mainly from bending moments and internal pressure, S_y is the hoop stress mainly from internal pressure, and τ_{xy} is mainly due to the torsion moment. Stresses due to direct axial and shear forces are generally insignificant and are often neglected.

There are a couple of things to note in applying Eq. (2.19). One is the nature of the bending stress, which always presents both tension and compression at a given cross-section. The other is the plane that includes the diametrical direction (through the thickness direction) also has to be included. The normal stress in this diametrical direction is either zero or equal to the acting pressure, which is compressive.

2.6.5 Effective Stress (von Mises Stress)

The maximum distortion energy failure theory agrees with the nature of ductile materials the most. This theory is very popular in the European piping community. Based on distortion energy theory, the condition for yielding is [7]

$$(S_1 - S_2)^2 + (S_2 - S_3)^2 + (S_1 - S_3)^2 = 2S_{yp}^2$$

where S_{yp} is the yield point tensile stress in the tension test. To make a stress directly comparable to the yield point stress and other data obtained in the tension test, the effective stress is coined and defined as

$$S_e = \sqrt{\frac{1}{2}[(S_1 - S_2)^2 + (S_2 - S_3)^2 + (S_1 - S_3)^2]} \tag{2.20}$$

For a two-dimensional stress field, as is commonly the case in practical piping system, S_3 can be set to zero, and we have

$$S_e = \sqrt{S_1^2 - S_1 S_2 + S_2^2}$$

The principal stresses S_1 and S_2 can be found using Eqs. (2.17a) and (2.17b), respectively, for two-dimensional stress systems. Substituting Eqs. (2.17a) and (2.17b) to the above, the effective stress for two-dimensional stress field becomes

$$S_e = \sqrt{(S_x - S_y)^2 + S_x S_y + 3\tau_{xy}^2} \tag{2.21}$$

By comparing Eq. (2.21) with Eq. (2.19), it is difficult to assess whether the effective stress is larger or smaller than the stress intensity. In other words, it is not possible to say which theory is more conservative. However, one thing is clear: when either one of the normal stresses is zero, the stress intensity is somewhat larger than the effective stress. The degree of difference depends on the ratio of the normal stress and the shear stress. At the extreme condition when both normal stresses are zero, the stress intensity is bigger than the effective stress by a factor of $\sqrt{4/3} = 1.155$. However, in most practical piping applications, both stress intensity and effective stress can be considered equivalent to each other.

2.7 BASIC BEAM FORMULAS

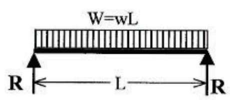
A piping system is essentially a group of beams connected together to form the shape required for transporting fluids from one point to another. Therefore, the behavior of the beam is the basic component of the pipe stress analysis. Table 2.1 shows some basic beam formulas that stress engineers are all familiar with. The following are a few important notes derived from these basic formulas:

(1) With a given loading, the bending moment is proportional to the length of the beam, but the displacement is proportional to the cube of the length. A slight increase in length creates a large increase in displacement, which translates to a large increase in flexibility.

(2) For a given configuration, the displacement is inversely proportional to EI. This EI is generally referred to as the *stiffness coefficient* of the cross-section. If a simulation is required for a non-standard cross-section such as refractory lined or concrete lined pipe, the EI of the simulating pipe has to match the combined EI of the pipes being simulated.

TABLE 2.1
BASIC BEAM FORMULAS

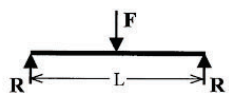
(a) Simple Beam – Uniform Load



$R = \frac{wL}{2}$, $M_{max} = \frac{wL^2}{8}$, (at Center)

$\Delta_{max} = \frac{5wL^4}{384EI}$, (at Center)

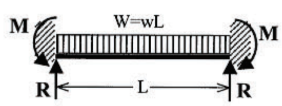
(b) Simple Beam – Concentrated Load



$R = \frac{F}{2}$, $M_{max} = \frac{FL}{4}$, (at Center)

$\Delta_{max} = \frac{FL^3}{48EI}$, (at Center)

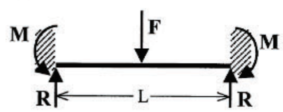
(c) Fixed Beam – Uniform Load



$R = \frac{wL}{2}$, $M_{max} = \frac{wL^2}{12}$, (at Ends)

$\Delta_{max} = \frac{wL^4}{384EI}$, (at Center)

(d) Fixed Beam – Concentrated Load



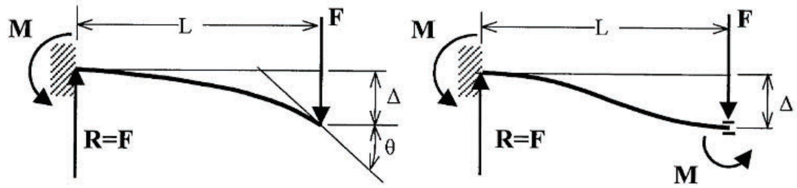
$R = \frac{F}{2}$, $M_{max} = \frac{FL}{8}$, (at Ends and Center)

$\Delta_{max} = \frac{FL^3}{192EI}$, (at Center)

(3) The displacement formulas include only the term EI. The terms involving shear modulus G and cross-section area A, associated with shear deformation, are not included. The formulas are good for practical lengths of beams. However, they are not accurate for short beams whose length is shorter than ten times the cross-sectional dimension of the beam. The formulas used by most computer programs include the shear deformation, thus making a short beam more flexible than that calculated by the formulas in Table 2.1. This is one of the potential discrepancies between a computer result and a hand calculation result. In such cases, the computer result is more accurate.

(4) Items (a) and (c) can be used to determine the stress due to weight under normal supporting spans. Because the actual piping in the plant is supported somewhere between simple support and fixed support [8], the following average formulas are generally used, instead, for evaluating weight supports.

$$S = \frac{M}{Z} = \frac{wL^2}{10Z}, \qquad \Delta = \frac{3wL^4}{384EI}$$



$$\Delta = \frac{FL^3}{3EI}, \quad F = \frac{3EI}{L^3}\Delta;$$

$$\theta = \frac{FL^2}{2EI}, \quad M = FL = \frac{3EI}{L^2}\Delta;$$

(a) Regular Cantilever Beam

$$\Delta = \frac{FL^3}{12EI}, \quad F = \frac{12EI}{L^3}\Delta$$

$$\theta = 0, \quad M = \frac{FL}{2} = \frac{6EI}{L^2}\Delta$$

(b) Guided Cantilever Beam

FIG. 2.14
DIFFERENT TYPES OF CANTILEVER BEAMS

2.7.1 Guided Cantilever

In piping flexibility analysis, the guided cantilever method is one of the often-mentioned approximate approaches [8]. Figure 2.14 shows two different types of cantilever beams that may be used in piping stress analysis. The regular cantilever beam shown in Figure 2.14(a) is actually half of the simple beam subject to a concentrated load as given in Table 2.1(b). The fixed end is equivalent to the mid-point of the simple beam. The guided cantilever beam shown in Figure 2.14(b) is actually half of the fixed beam subject to a concentrated load as given in Table 2.1(d). These beam models are often used to calculate the forces and moments, approximately, in a given length of pipe subject to a displacement resulting from thermal expansion.

The regular cantilever model requires an accompanying rotation, θ, while absorbing the displacement, Δ. This rotational relief reduces the forces and moments generated by a given displacement. The problem is that an actual piping system does not freely offer this rotational relief. It does offer some relief, but much less than the amount shown in Fig. 2.14(a). A more realistic and conservative approach is to assume that no rotation is taking place, as shown in Fig. 2.14(b). This is the so-called guided cantilever approach. For a given expansion or displacement, the force created by the guided cantilever is four times as great as that by a regular cantilever, and the moment, thus the stress is twice as large as that by a regular cantilever.

2.8 ANALYSIS OF PIPING ASSEMBLY

The analysis of a piping assembly is very complicated and is accomplished mostly by using computer programs. In general, an engineer is not required to have knowledge of the computing methods implemented in the computer programs in order to use the program. However, some common sense regarding general analytical approaches can help analysts to better understand the procedure and better interpret the results.

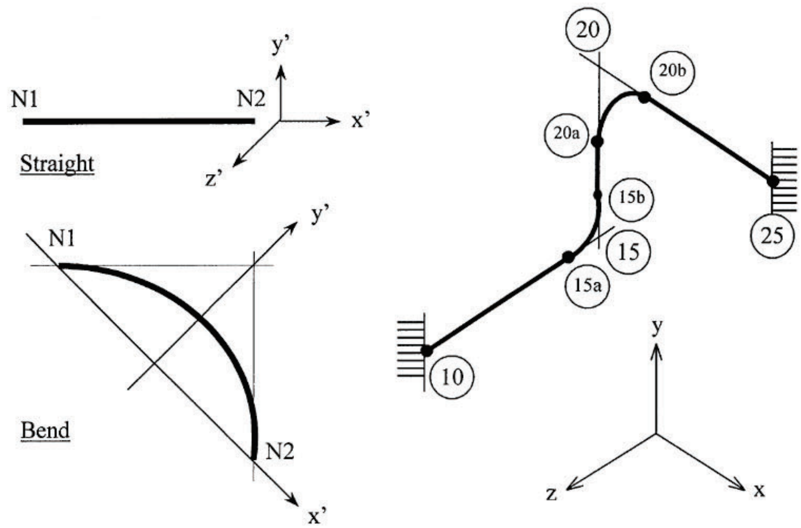
A piping system consists of many components laid out in all directions. Analysis of the piping system is normally idealized into a combination of straight pipe segments and pipe bend elements. In other words, the analysis is performed using a finite element method consisting of two types of elements: the straight pipe and the pipe bend.

2.8.1 Finite Element

The body of a structure generally has stresses and strains that vary continuously throughout the body. It is very difficult, if not impossible, to calculate exactly these stresses and strains. However, to obtain a practical result, the body can be divided into many sub-bodies, each with a finite size. Each small body is considered to have a predictable stress and strain distribution over it. These small bodies are called finite elements. In piping analysis, these bodies are actually fairly large compared to the general sense of a finite element. Here, we use straight pipe and curved pipe, two types of beam elements. Each element has two nodes, N1 and N2, as shown in Fig. 2.15(a). N1 is the beginning node, and N2 is the ending node.

The characteristics of the element are expressed in the local coordinates aligned with the element geometry. For a straight pipe element, the local *x*-axis is always in the axial direction pointing from the beginning node toward the ending node. The local *y*-axis and *z*-axis are perpendicular to each other in the lateral directions. For a curved pipe element, the local axis convention differs slightly among different investigators. One popular convention, as shown in Fig. 2.15(a), assigns the local *x*-axis as connecting the two nodes and pointing from the beginning node to the ending node. The local *y*-axis is perpendicular to the *x*-axis and pointing from the mid-cord point toward the bend tangent intersection point.

In a general three-dimensional environment, each node has six degrees of freedom, three in translation and three in rotation. At each element, the forces and displacements have a fixed relationship in local coordinates and are designated with a prime (′) notation. Each node of an element is associated with three displacements, D_x', D_y', D_z', and three rotations, R_x', R_y', R_z'. Correspondingly, each node is also associated with three forces, F_x', F_y', F_z', and three moments, M_x', M_y', M_z'. In general, the term displacement is used to cover both displacement and rotation. By the same token, the term force is



(a) Local Coordinates (b) Global Coordinates & Assembly

FIG. 2.15
COORDINATE SYSTEMS

used to cover both forces and moments. The finite element method is built under the premise that on each element there is a relationship between these forces and displacements. That is, for each element we have the relation in local coordinates as

$$\{\mathbf{F}'\} = [K']\{\mathbf{D}'\} \tag{2.22}$$

where $\{\mathbf{F}'\}$ is the force vector representing 12 forces and moments at both nodes. That is, $\{\mathbf{F}'\} = \{F'_{x_1}, F'_{y_1}, F'_{z_1}, M'_{x_1}, M'_{y_1}, M'_{z_1}, F'_{x_2}, F'_{y_2}, F'_{z_2}, M'_{x_2}, M'_{y_2}, M'_{z_2}\}^{\mathrm{T}}$. Subscript 1 represents node N1 and subscript 2 represents node N2. Superscript T denotes transpose, meaning a column vector. $\{\mathbf{D}'\}$ is the displacement vector representing 12 displacements and rotations at both nodes. That is $\{\mathbf{D}'\} = \{D'_{x_1}, D'_{y_1}, D'_{z_1}, R'_{x_1}, R'_{y_1}, R'_{z_1}, D'_{x_2}, D'_{y_2}, D'_{z_2}, R'_{x_2}, R'_{y_2}, R'_{z_2}\}^{\mathrm{T}}$. $[K']$ is a 12×12 symmetric stiffness matrix. The exact terms in $[K']$ are too complex to be included here. Interested readers may find them in related publications [9–12]. There are several different forms of these terms, using somewhat different local coordinate conventions.

2.8.2 Data Points and Node Points

Before a piping system is analyzed, every element in the system has to be identified. Customarily, these elements are identified with point numbers. These numbers serve as the communication addresses just like real-life house numbers. Up to three different point numbers may be associated with an analysis. Take, for example, the simple assembly shown in Fig. 2.15(b). The first set of numbers needed is the set of data point numbers used to describe the geometry of the system. The following are the locations that need to be assigned with data point numbers:

(1) Terminal points such as anchors, free ends, vessel, and tank connections, etc. Points 10 and 25 belong to this category.
(2) Bend tangent intersection points (working points). Points 15 and 20 belong to this category.
(3) Branch intersection points.
(4) Key flange face points.
(5) Restraint and loading points.
(6) Other points where the response of the system is of interest.

The above data points are required for a precise description of the system. From these essential data points, the computer program may also generate some other data points required for the analysis. At bend 15, for instance, the required input data point is the tangent intersection point 15, but the points required for the analysis are the end points 15a and 15b of the bend. Point 15 is not located at any physical part of the piping system; therefore, it is not used in the analysis.

From the information provided by the data points, the computer program will re-number the entire piping system to assign analysis node numbers. The analysis node numbers have to be consecutive starting from 1. They are generated following the input sequence of the data. In our example system, we have nodes 1 = 10, 2 = 15a, 3 = 15b, and so forth. In an actual analysis, these sequentially generated node numbers may be once again re-numbered to achieve the so-called bandwidth optimization. The bandwidth is essentially the difference between the node number of a given point and the node number of its adjacent connecting point or points. For a multi-branched system, the bandwidth greatly depends on the re-numbering. The result of this re-numbering is another set of node numbers, which are called re-numbered node numbers. The last re-numbering is intended to achieve the minimum bandwidth possible. Smaller bandwidths require less data storage and also less computing time. The whole numbering effort may appear very confusing, but it should not be a concern of the analyst. All analysis results are given in reference to the original data point numbers assigned.

2.8.3 Piping Assembly

To mathematically assemble the piping system, all individual elements have to use a common coordinate system. This common coordinate system is called the global coordinate system. The global *y*-axis is generally fixed in the vertical direction pointing upward. This practice is partly due to piping tradition [8] and partly for the convenience of specifying the gravitational load. This *y* coordinate assignment also makes it easier to correlate it with the elevation that is often used in a piping system. The global *x*- and *z*-axes are normally aligned with plant's major orientations, which are often called plant-North and plant-East, etc.

The local coordinate stiffness matrix of each element has to be first converted to global coordinates before a mathematical assembly can be performed. Depending on the orientation of the element, a rotational matrix consisting of directional parameters is used to convert the local force and local displacement vectors into global force and global displacement vectors. That is,

$$\{\mathbf{F}'\} = [L]\{\mathbf{F}\}, \qquad \{\mathbf{D}'\} = [L]\{\mathbf{D}\}$$

where

$[L]$ = 12×12 transformation matrix consisting of four sets of 3×3 rotational parameter matrices placed at the diagonal
$\{\mathbf{F}\}$ = force vector in global coordinates
$\{\mathrm{D}\}$ = displacement vector in global coordinates

Substituting the above relations to Eq. (2.22), we have

$$[L]\{\mathbf{F}\} = [K'][L]\{\mathbf{D}\} \qquad \text{or} \qquad \{\mathbf{F}\} = [L]^{-1}[K'][L]\{\mathbf{D}\}$$

The global stiffness matrix, $[K]$, of each element is then created by applying the rotational transformation on the local stiffness matrix $[K']$ as

$$\{\mathbf{F}\} = [K]\{D\}, \qquad \text{where } [K] = [L]^{-1}[K'][L] = [L]^{\mathrm{T}}[K'][L] \tag{2.23}$$

$[L]^{-1}$ is the inverse of the transformation matrix $[L]$, and $[L]^{\mathrm{T}}$ is the transposition of $[L]$. For rotational transformation matrices, $[L]^{-1}$ and $[L]^{\mathrm{T}}$ are equal. The transposition is obtained by interchanging the rows and columns of the original matrix, $[L]$. In the actual assembling process, the force/displacement relation given in Eq. (2.23) is partitioned to

$$\begin{vmatrix} F_1 \\ F_2 \end{vmatrix} = \begin{vmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{vmatrix} \begin{vmatrix} D_1 \\ D_2 \end{vmatrix} \tag{2.24}$$

where subscripts 1 and 2 denote node N1 and node N2, respectively. Each F_* and D_* sub-vector represents six components corresponding to six degrees of freedom.

For a system with *n*-node points, the total number of degrees of freedom is $N = 6n$. These N degrees of freedom are all potentially related. Therefore, the system has $(N \times N)$ overall stiffness matrix relating the forces at all degrees of freedom with the displacements at all degrees of freedom. The assembly of the overall stiffness matrix uses the force vector and displacement vector, both representing all degrees of freedom, as the starting template. Each element matrix as given by Eq. (2.24) is placed inside the overall stiffness matrix at the location representing the proper degree of freedom in the force and displacement template. All element stiffness values placed at the same location inside the overall stiffness are simply added together algebraically. The assembly is completed when all element matrices are included in the overall matrix. The overall matrix of a piping system generally forms a narrow band around the diagonal. The width of this band is called the bandwidth of the matrix. Again, because a smaller bandwidth requires less storage and expedites calculation, it is generally desirable to reduce the bandwidth by re-numbering the node numbers. The assembly of the overall matrix is based on the renumbered nodes. Restraints and anchors are treated as additional stiffness added to the diagonal of the corresponding node location. The final overall equation looks like

$$\begin{vmatrix} F_1 \\ F_2 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ F_N \end{vmatrix} = \begin{vmatrix} K_{11} & K_{12} & \cdot & \cdot & \cdot & \cdot & K_{1N} \\ K_{21} & K_{22} & K_{23} & \cdot & \cdot & \cdot & K_{2N} \\ \cdot & \cdot & K_{33} & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ K_{N1} & \cdot & \cdot & \cdot & \cdot & \cdot & K_{NN} \end{vmatrix} \begin{vmatrix} D_1 \\ D_2 \\ \cdot \\ \cdot \\ \cdot \\ \cdot \\ D_N \end{vmatrix} \tag{2.25}$$

where N is the total number of degrees of freedom, which is equal to six times the number of node points. This is also often referred to as the number of equations.

For the example system shown in Fig. 2.15(b), there are six node points. Therefore, the force vector and displacement vector each has 36 components. The size of the overall stiffness matrix is 36×36. The displacement vector is the unknown quantity, but the force vector is associated with the load and is considered as the known quantity. Therefore, the analysis is reduced to the problem of solving for the displacements at all node points based on the known forces at all node points. This is the equivalent of solving $6n$ simultaneous linear equations defined by Eq. (2.25). The discussion of the solution technique is beyond the scope of this book. Interested readers can read, for instance, the book by Bathe and Wilson [12].

After the nodal displacements are solved for each load case, the forces and moments (in global coordinates) at each element can be found by using Eq. (2.24). These global forces and moments have to be transformed back to the local coordinate before the stresses can be calculated.

REFERENCES

[1] American Society for Testing and Materials (ASTM), "Standard Methods and Definitions for Mechanical Testing of Steel Products," ASTM A-370.

[2] American Society of Mechanical Engineers (ASME), *ASME Code for Pressure Piping, B31, An American National Standard*, ASME, New York.

[3] European Standard EN 13480-3, 2002, *Metallic Industrial Piping — Part 3: Design and calculation*, CEN European Committee for Standardization.

[4] API Spec 5L, *API Specification for Line Pipe*, American Petroleum Institute, Washington, D.C.

[5] Young, W. C., 1989, *Roark's Formulas for Stress & Strain*, 6th ed., McGraw-Hill Book Company, New York.

[6] American Institute of Steel Construction, Inc., *Manual of Steel Construction*, AISC, Chicago.

[7] Timoshenko, S., 1956, *Strength of Materials, Part-II*, 3rd ed., Van Nostrand Co., New York.

[8] M. W. Kellogg Company, 1956, *Design of Piping Systems*, revised 2nd ed., John Wiley & Sons, New York.

[9] Przemieniecki, J. S., 1968, *Theory of Matrix Structural Analysis*, McGraw-Hill Book Company, New York.

[10] Brock, J. E., 1952, "A Matrix Analysis for Flexibility Analysis of Piping Systems," *Journal of Applied Mechanics*, *19*(4), pp. 501–516.

[11] Brock, J. E., 1955, "Matrix Analysis of Piping Flexibility," *Journal of Applied Mechanics*, *22*(3), pp. 361–362.

[12] Bathe, K.-J., and Wilson, E. L., 1976, *Numerical Methods in Finite Element Analysis*, Prentice-Hall, Englewood Cliffs, NJ.